

Decision Tree

→ supervised

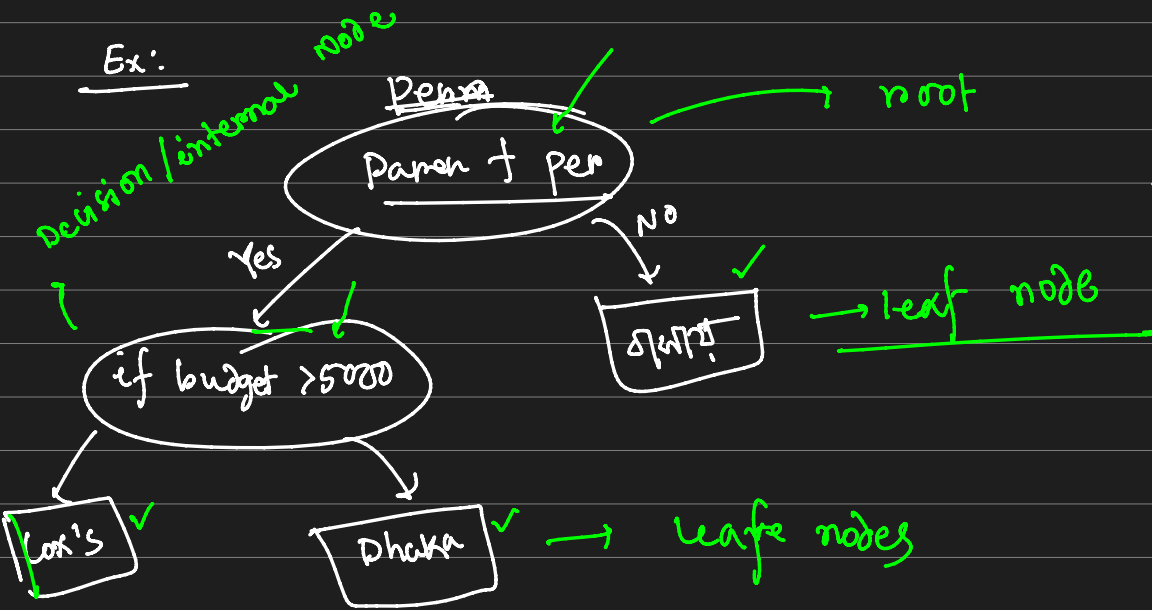
Classification xx

- ① Logistic → eq
- ② KNN → distance

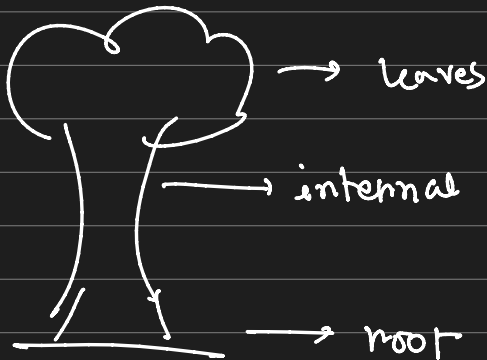
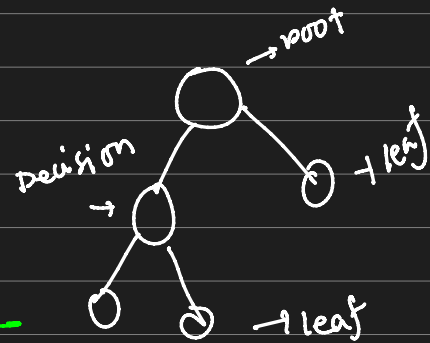
rules based

Decision

Ex:



Tree

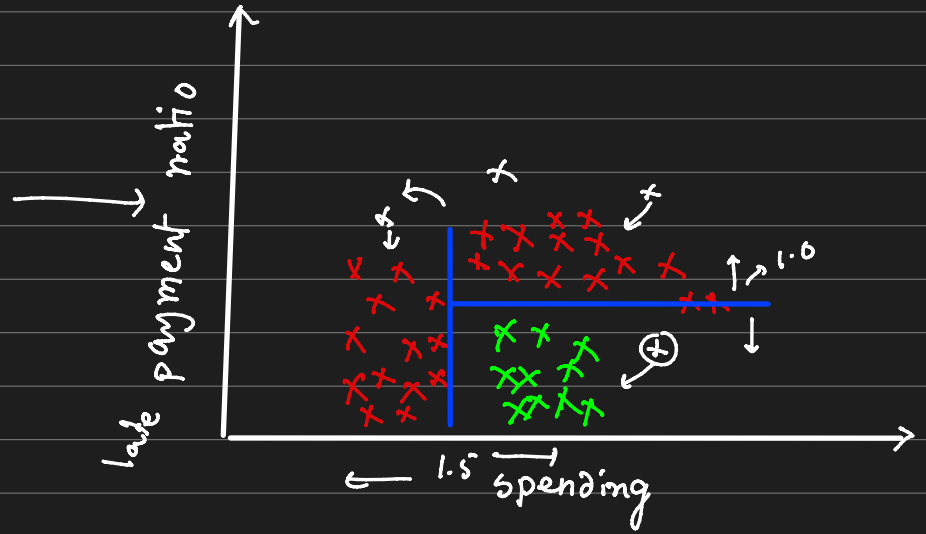


for 9/21/21

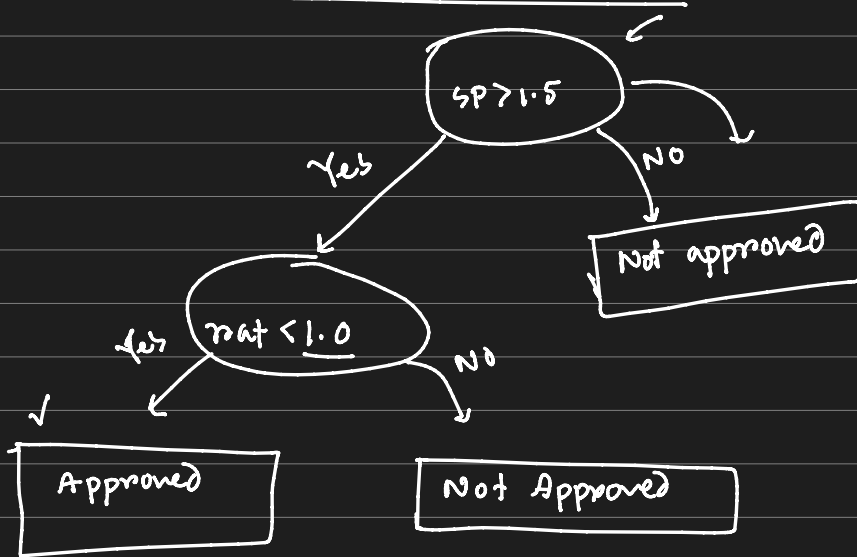
Decision Tree on a Dataset

loan Approval:

SP	rat	App
1.2	0.5	No
1.3	0.7	No
1.5	1.2	No
1.8	0.5	Yes
1.9	0.6	Yes
1.9	1.2	No
1.9	1.5	No

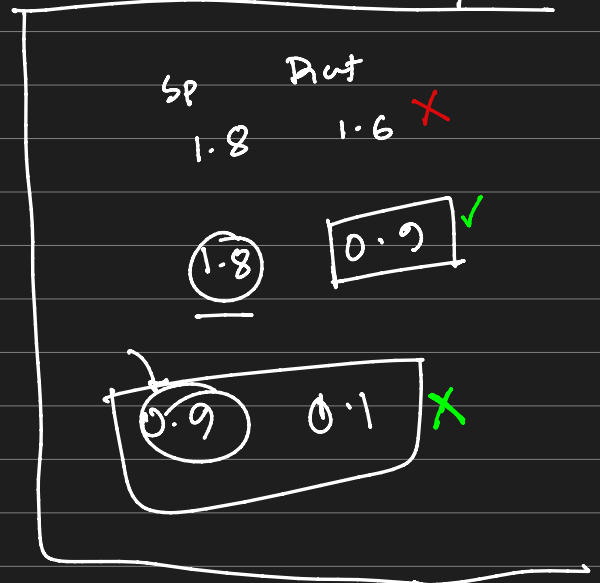


Decision Tree

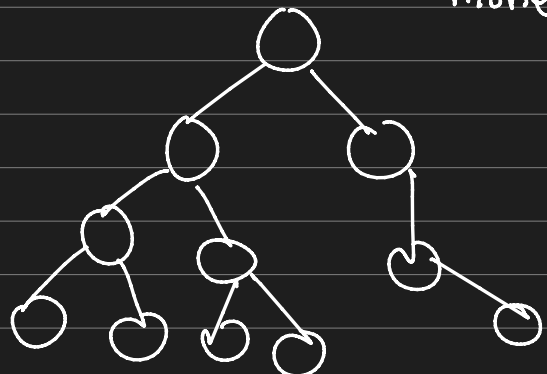


if feat 50 used

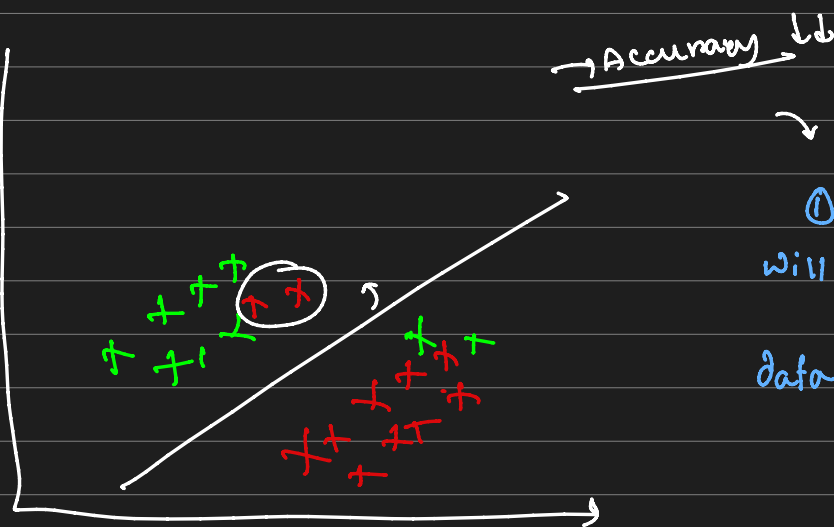
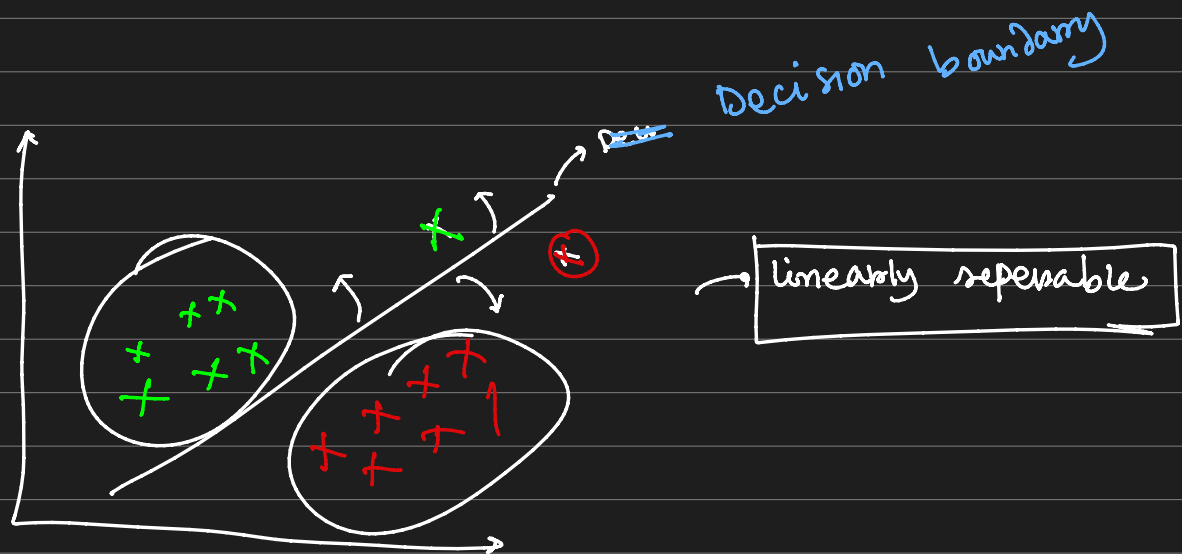
points to predict



A random tree can be more complex



Logistic Regression Decision Boundary



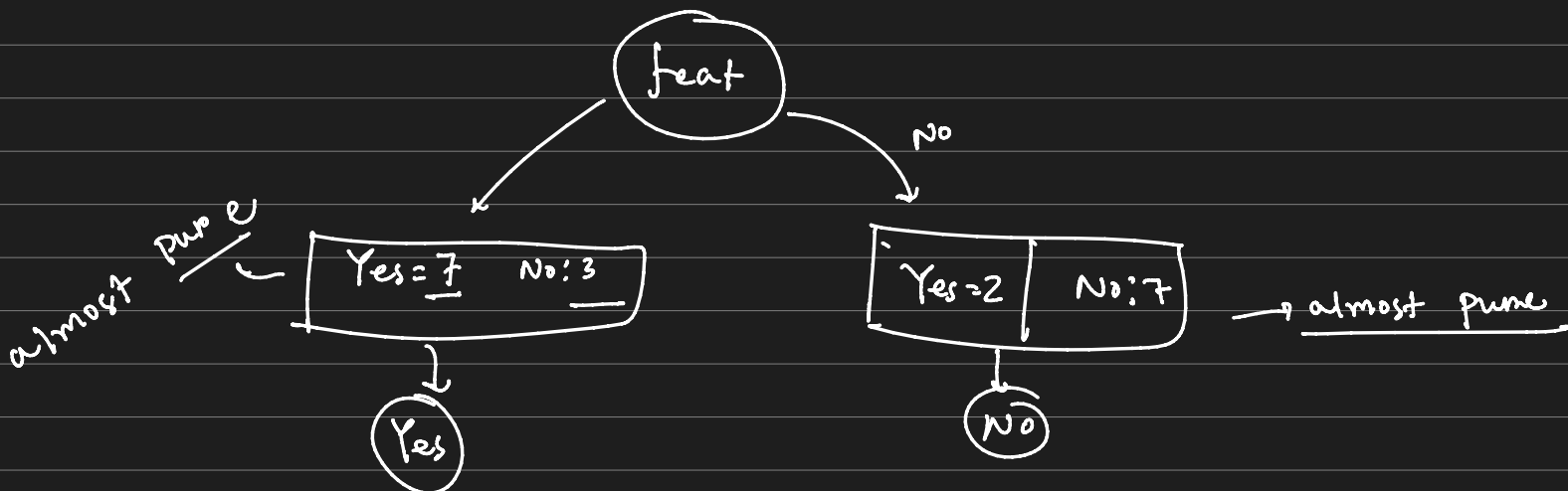
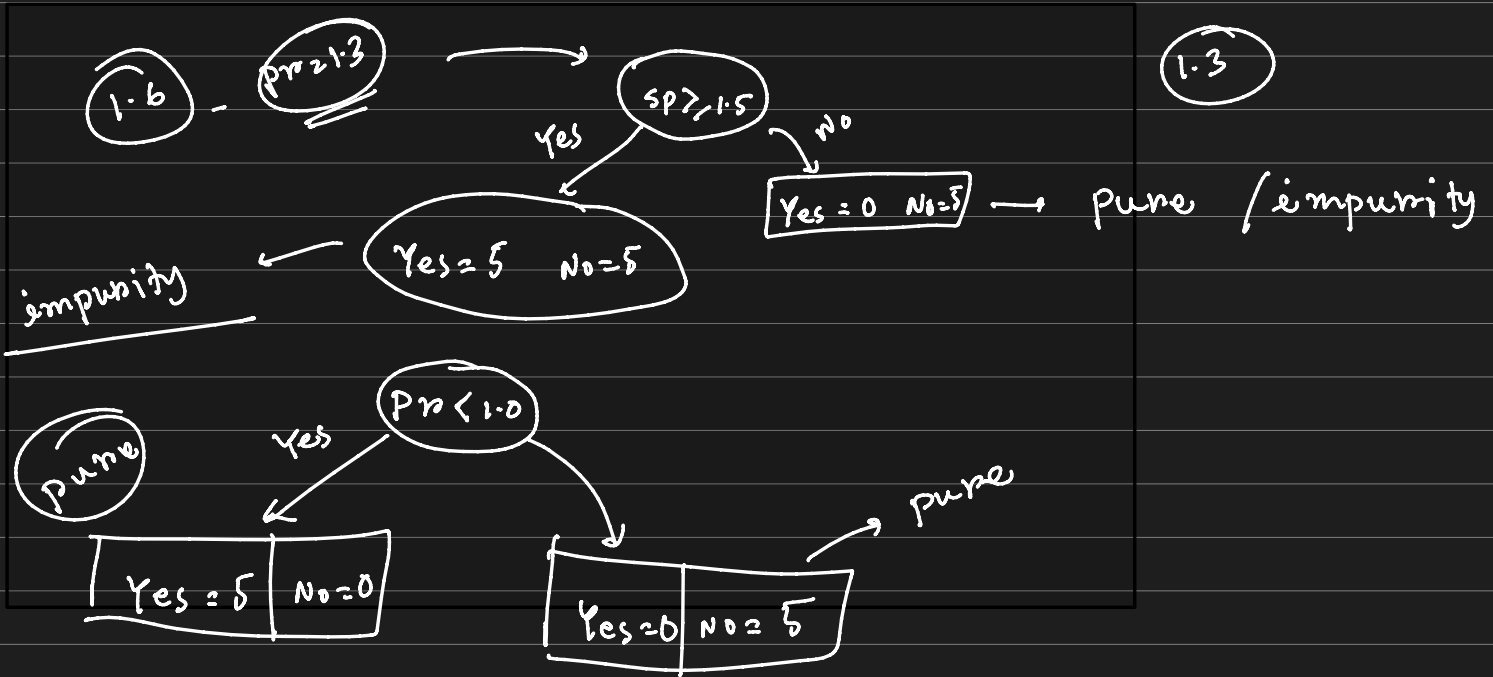
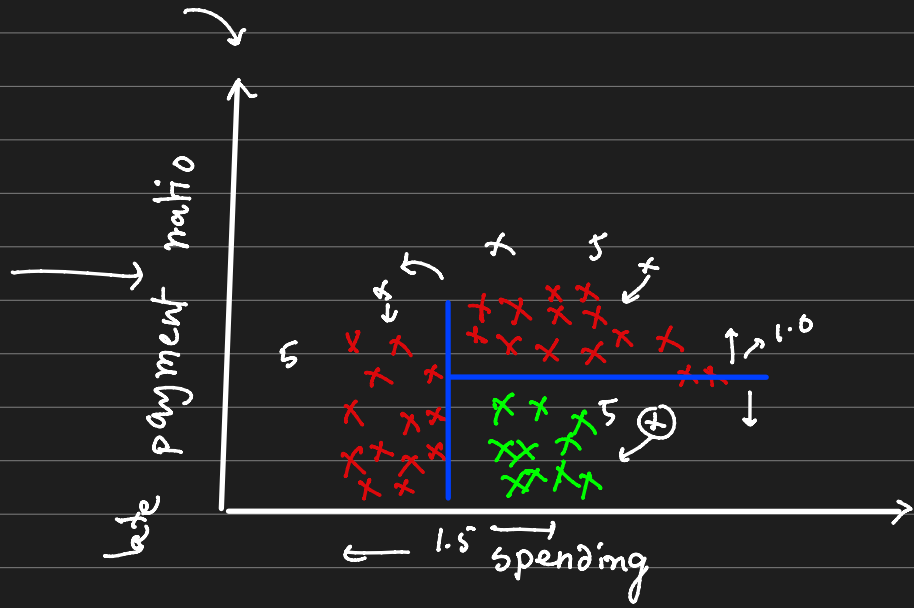
① That means accuracy will be too low if data is non-linear.

Almost linearly separable

Metrics to choose features

loan Approval:

SP	rat	App
1.2	0.5	No
1.3	0.7	No
1.5	1.2	No
1.8	0.8	Yes
1.9	0.6	Yes
1.8	1.2	No
1.7	1.5	No

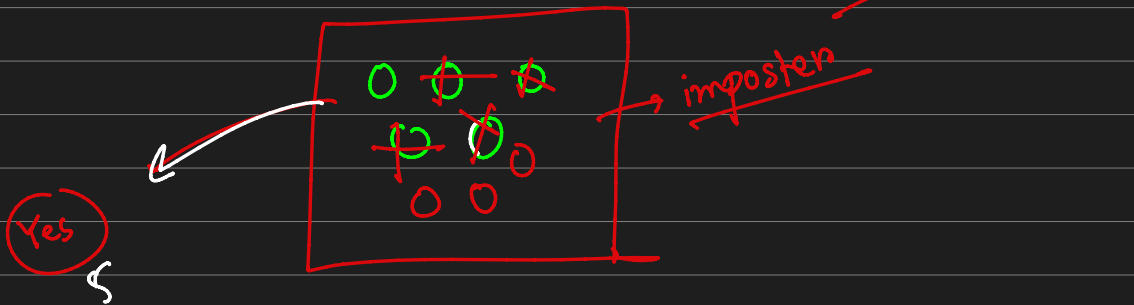


Metrics: →

- (i) entropy
- (ii) Gini Impurity
- (iii) Information Gain

Entropy → अशुद्धता

(i) Among Us



(ii) Yes No

$$\text{Entropy} = \sum_{i=1}^n -p_i \log_2(p_i)$$

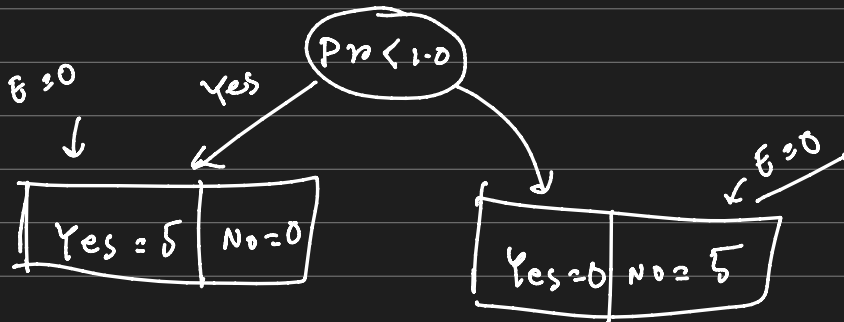
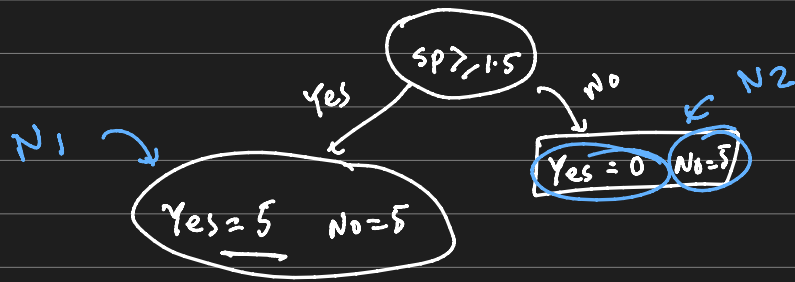
n → total class

Yes } → 2 class
No }

$p_i = i^{\text{th}}$ class ka probability.

Cat } → 3 class
Dog }
Rabbit }

$$\text{Prob} = \frac{\text{Particular class ka}}{\text{total ka}}$$



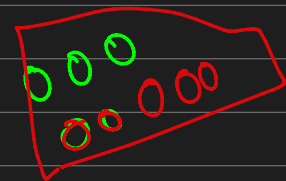
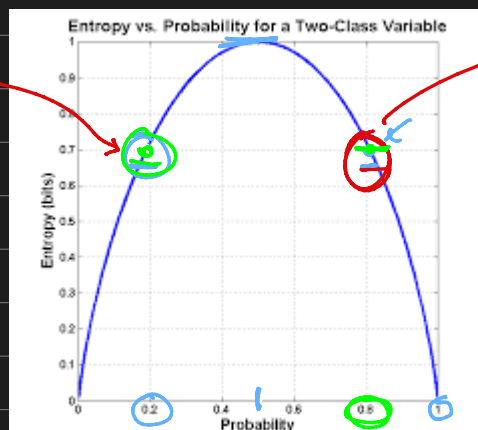
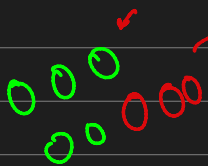
$$E(N_1) = - \frac{5}{10} \log\left(\frac{5}{10}\right) - \frac{5}{10} \log\left(\frac{5}{10}\right)$$

$$= 1.0$$

$$E(N_2) = - \frac{5}{5} \log\left(\frac{5}{5}\right)$$

$$= -1 \times \log(1)$$

$$= -1 \times 0 = 0$$



50%
50%

$Yes = 8$ $No = 2$

$\frac{8}{10}$

$Yes = 0.8$
 $No = 0.2$

$Yes = 0.2$
 $No = 0.8$

$1 - 0.2 = 0.8$

Max Impurity
 $Y=5$
 $N=5 \rightarrow 10$

Gini Impurity

$$1 - (P_A^2 + P_B^2 + P_C^2)$$

$$G_I = 1 - (P_Y^2 + P_N^2)$$

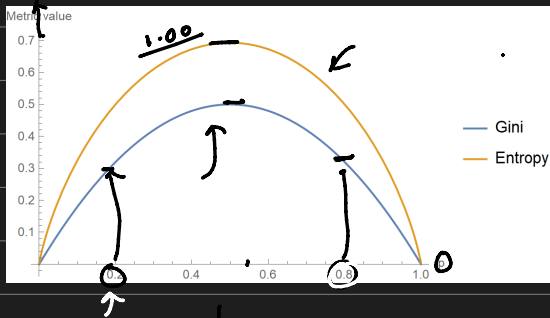
$$= 1 - \left\{ \left(\frac{5}{10} \right)^2 + \left(\frac{5}{10} \right)^2 \right\} = 0.5$$

Pure
 $Y=0, N=5$

$$G_I = 1 - \left\{ 0^2 + \left(\frac{5}{5} \right)^2 \right\}$$

$$= 1 - (0 + 1)$$

$$= 1 - 1 = 0$$



$$Y=8, N=2 \begin{cases} P_Y = \frac{8}{10} = 0.8 \\ P_N = 1 - 0.8 = 0.2 \end{cases}$$

$$Y=2, N=8 \begin{cases} P_Y = \frac{2}{10} = 0.2 \\ P_N = 1 - 0.2 = 0.8 \end{cases}$$

① For dataset 1 Gini Impurity better

$$Y=5, N=5$$

DP information

$$Y=8, N=2$$

How to choose?
 tuning
 Hyperparameter tuning.

Information Gain

କଟା feature ଥିବା

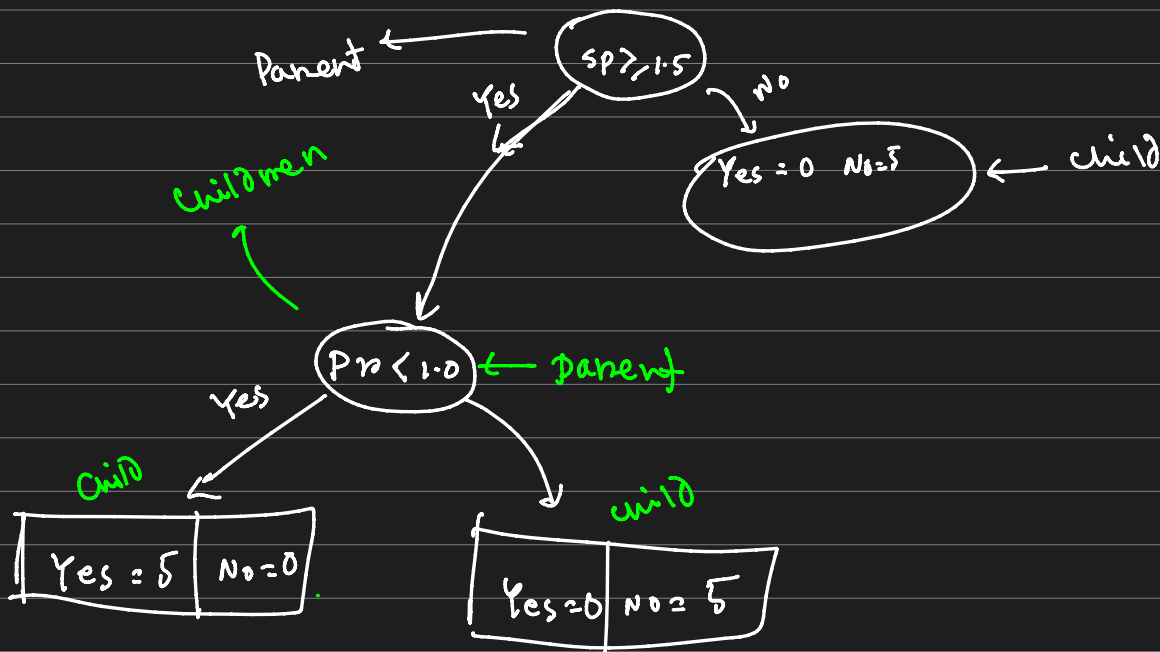
କଟା split

Procedure of Decision Tree:

- (i) Information Gain Calculation
- (ii) Best information gain
- (iii) ଥିବା feat pick କଟା split
↳ pure / rules ଗଠିବ

Entropy, Gini

$$I_G = \underline{E(\text{Parent})} - \text{Weighted Av of } E(\text{children})$$



Dataset: ← ↘ Target

ENERGY	Weather	Go out
High	Sunny	YES
low	Sunny	No
low	Rainy	No
High	Rainy	YES

ST:

$E(\text{parent}) =$

$$- \left\{ \frac{2}{4} \log\left(\frac{2}{4}\right) + \frac{2}{4} \log\left(\frac{2}{4}\right) \right\}$$

$$= 1$$

Y=2
N=2

Energy

$E=0$

En	W	G
High	Sun	YES
High	Rain	YES

$E=0$

En	W	G
low	Sun	No
low	Rain	No

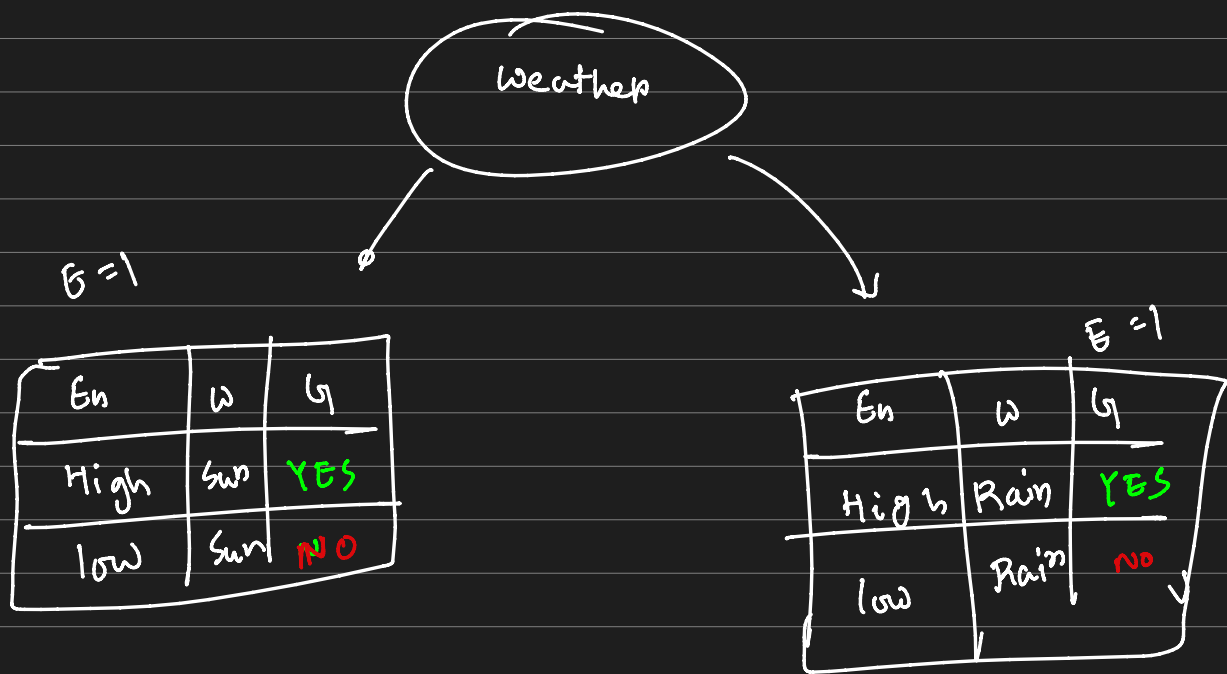
①

$$I_G = \underline{E(\text{Parent})} - \text{Weighted Av of } E(\text{children})$$

$$= 1 - \left\{ \frac{2}{4} \times 0 + \frac{2}{4} \times 0 \right\} = 1$$

weighted
Avg

$$E(\text{child}) = \frac{\text{children \& datapoint}}{\text{parent \& datapoint}} \times E(\text{child})$$



$$I_G = E(\text{parent}) - w \cdot \text{Avg } E(\text{children})$$

$$= 1 - \left\{ \frac{2}{4} \times 1 + \frac{2}{4} \times 1 \right\}$$

$$= 1 - \frac{4}{4}$$

$$= 1 - 1 = 0$$

Lost

building a decision tree from Scratch

Outlook	Temp	Routine	Wears lost
Sun	Cold	In	No
Sun	Warm	Out	No
Cloud	Warm	In	No
Sun	Warm	In	No
Cloud	Cold	In	YES
Cloud	Cold	Out	YES
Sun	Cold	Out	YES

No = 7
YES = 3

$$E(\text{parent}) = - \left\{ \frac{3}{10} \log_2 \left(\frac{3}{10} \right) + \frac{7}{10} \log_2 \left(\frac{7}{10} \right) \right\}$$

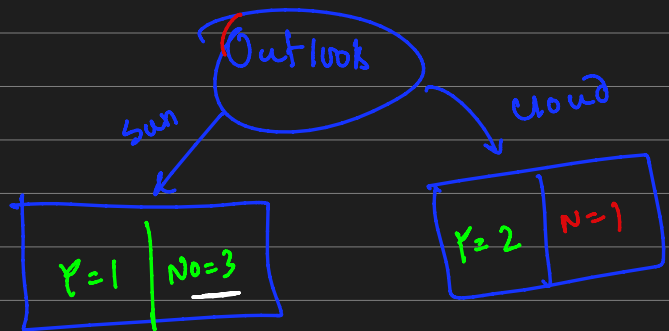
$$= 0.985$$

$$I_G = E(\text{parent}) - w \cdot \text{Avg } E(\text{children})$$

$$= 0.985$$

$$E(\text{parent}) = 0.985$$

Outlook:



$$E(\text{Out - Sunny}) = - \left\{ \frac{1}{4} \log_2 \left(\frac{1}{4} \right) + \frac{3}{4} \log_2 \left(\frac{3}{4} \right) \right\}$$

$$= \boxed{0.811}$$

$$E(\text{Out - Cloudy}) = - \left\{ \frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right\}$$

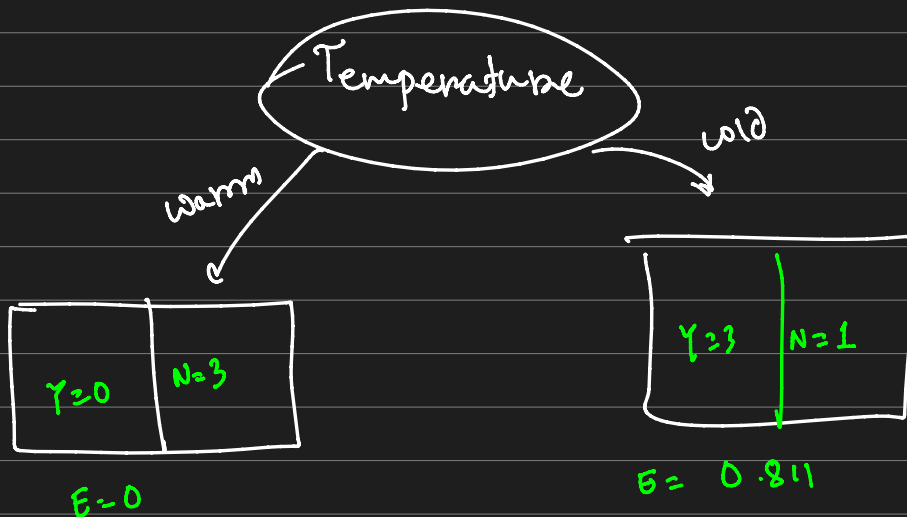
$\uparrow$ Yes: prob $\uparrow$ No: prob

$$= 0.918$$

$$\underline{I_G} = E(P) = \text{weighted Avg } E(\text{child})$$

$$= 0.985 = \left\{ \frac{4}{7} \times 0.811 + \frac{3}{7} \times 0.918 \right\}$$

$$= \underline{0.128}$$



$$I_G = E(P) = \text{w.Avg} \times E(y)$$

$$= 0.985 = \left\{ \frac{3}{7} \times 0 + \frac{4}{7} \times 0.811 \right\}$$

$$= \underline{0.521}$$

Out	$I_G = 0.128$
Temp	$I_G = 0.521$
Routine	$I_G = \underline{0.128}$



$Y=3, NO=1$



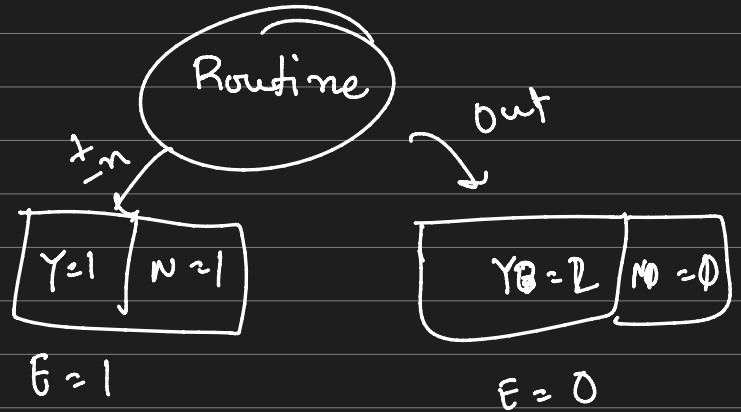
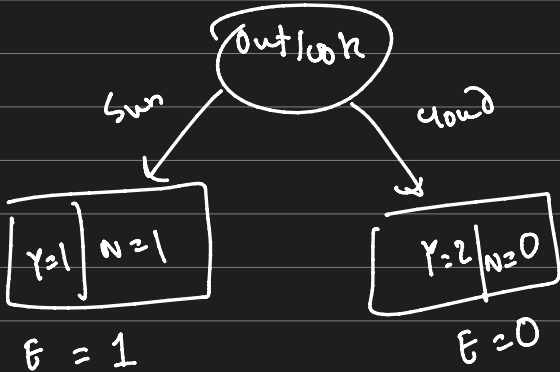
$Y=3, NO=1$

Outlook	Temp	Routine	Wears hat
Sun	Cold	In	No
Sun	Warm	Out	No
Cloud	Warm	In	No
Sun	Warm	In	No
Cloud	Cold	In	Yes
Cloud	Cold	Out	Yes
Sun	Cold	Out	Yes

$$E(\text{Parent}) = - \left\{ \frac{3}{4} \log \left(\frac{3}{4} \right) + \frac{1}{4} \log \frac{1}{4} \right\}$$

$$= 0.811$$

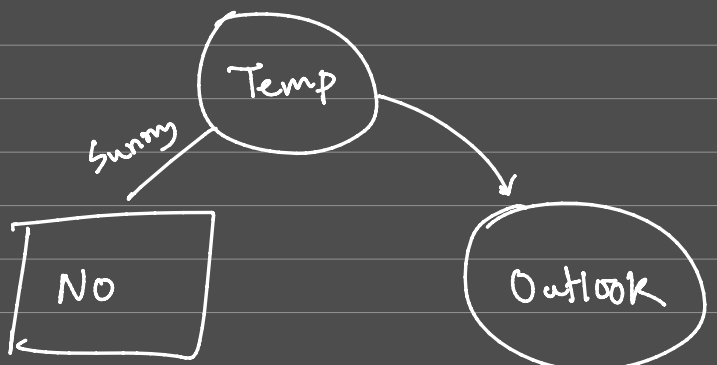
$$I_G = E(\text{Parent}) - \text{Avg Weighted Avg } E(\text{child})$$



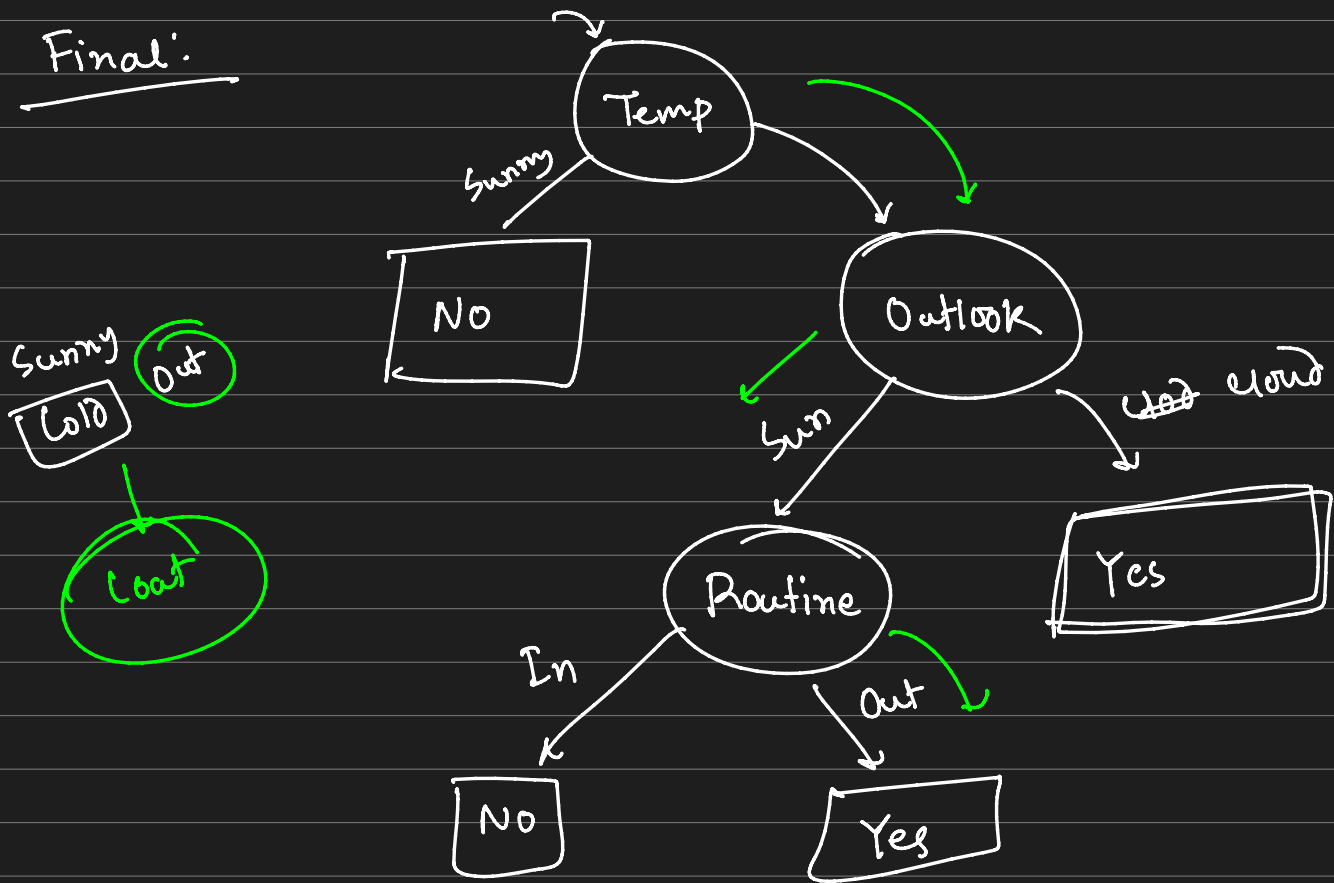
$$I_G = 0.811 - \left(\frac{2}{4} \times 1 - 0 \right)$$

$$= 0.311$$

$$I_G = 0.311$$



Final:



Outlook	Temp	Routine	Wear coat
Sun	Cold	In	No
Sun	Warm	Out	No
Cloud	Warm	In	No
Sun	Warm	In	No
Cloud	Cold	In	Yes
Cloud	Cold	Out	Yes
Sun	Cold	Out	Yes

Let's Code Decision Tree

~ (R)